# Diameter Growth Equations for CONUS with Climate Variables

Greg Johnson, David Marshall, Aaron Weiskittel

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## Fit Diameter Growth Equations (no site index)

### Preliminary CONUS-wide Diameter Growth Equation

We built a data set from Forest Inventory and Analysis (FIA) remeasurement data, filtering out observations with missing values for diameter at breast height (`dbh`), basal area in larger trees (`bal`), crown ratio (`cr`), total height (`ht`), site index (`si`), summer precipitation ($PPT_{sm}$), extreme minimum temperature (`EMT`), and remeasurement intervals less than 5 years. We also restricted the data to species with $\geq$ 5000 observations.

The diameter growth equation fit to these data was:

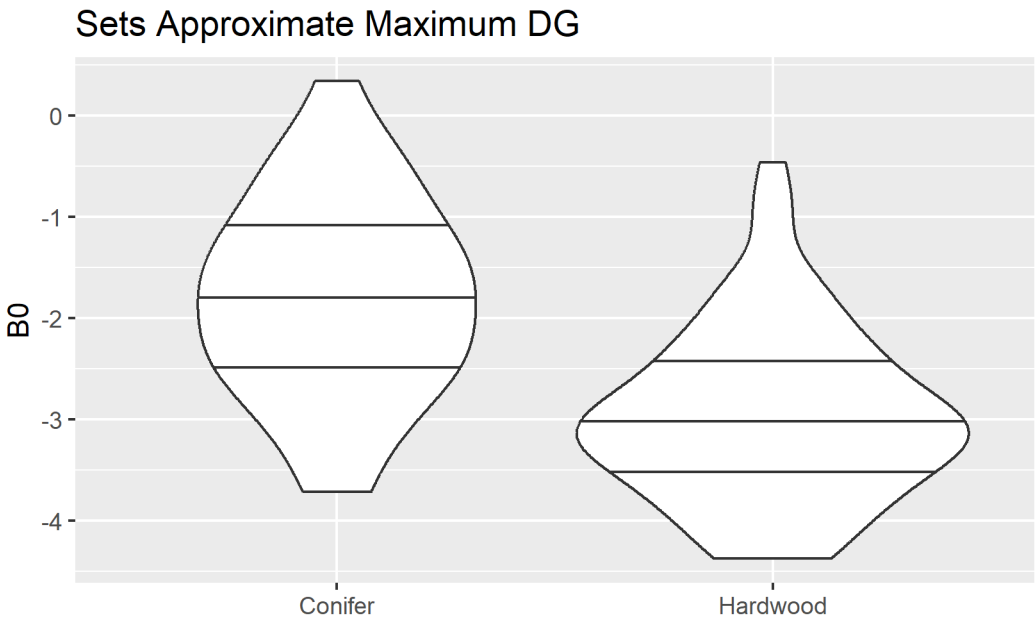
$$\Delta dbh = e^{(\beta_0+\beta_1 log(\frac{(dbh+1)^2}{(cr*ht+1)^{\beta_4}})+\beta_2 \frac{bal^{\beta_5}}{dbh+2.7}+\beta_3 PPT_{sm}+\beta_6 EMT)} \tag{1}$$

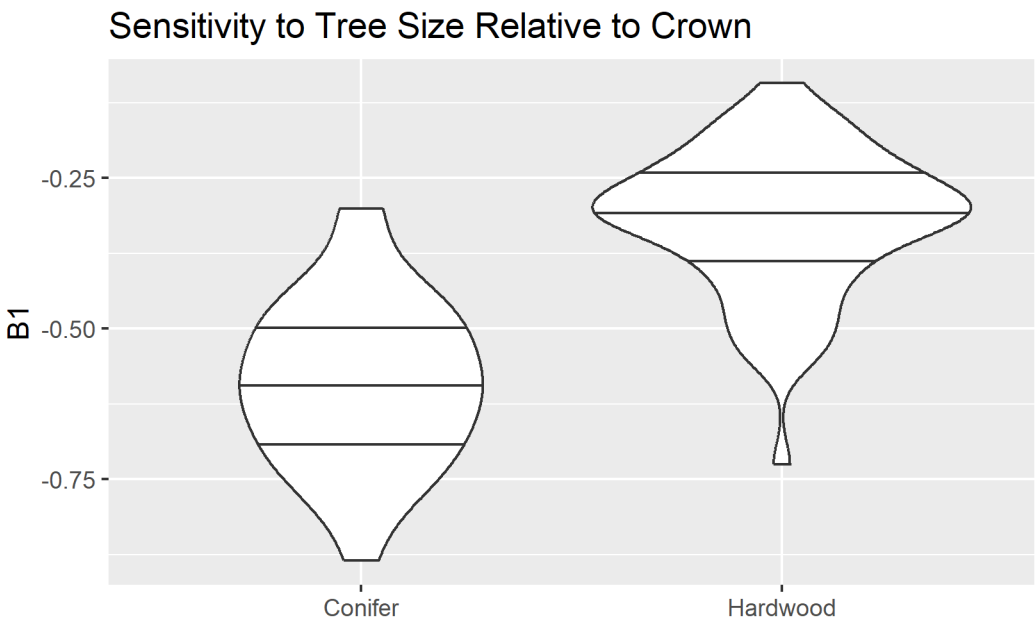
where:

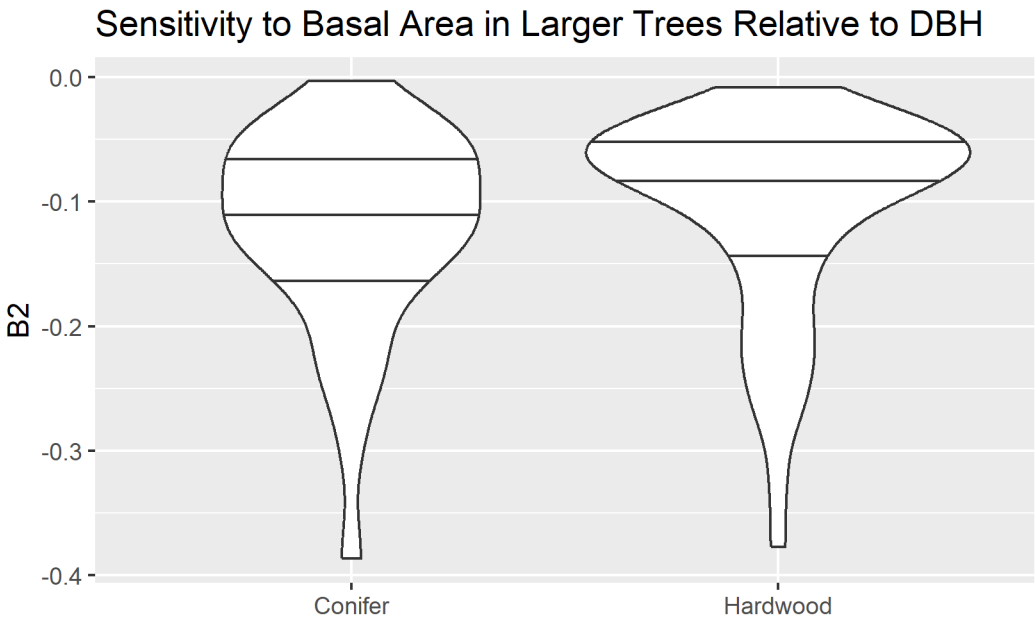
- $dbh$ = diameter at breast height (inches)
- $cr$ = crown ratio (fraction of total height)
- $ht$ = total height (feet)
- $bal$ = basal area in larger trees ($feet^2/acre$)
- $PPT_{sm}$ = summer precipitation (mm)
- $EMT$ = extreme minimum temperature ($^\circ C$)

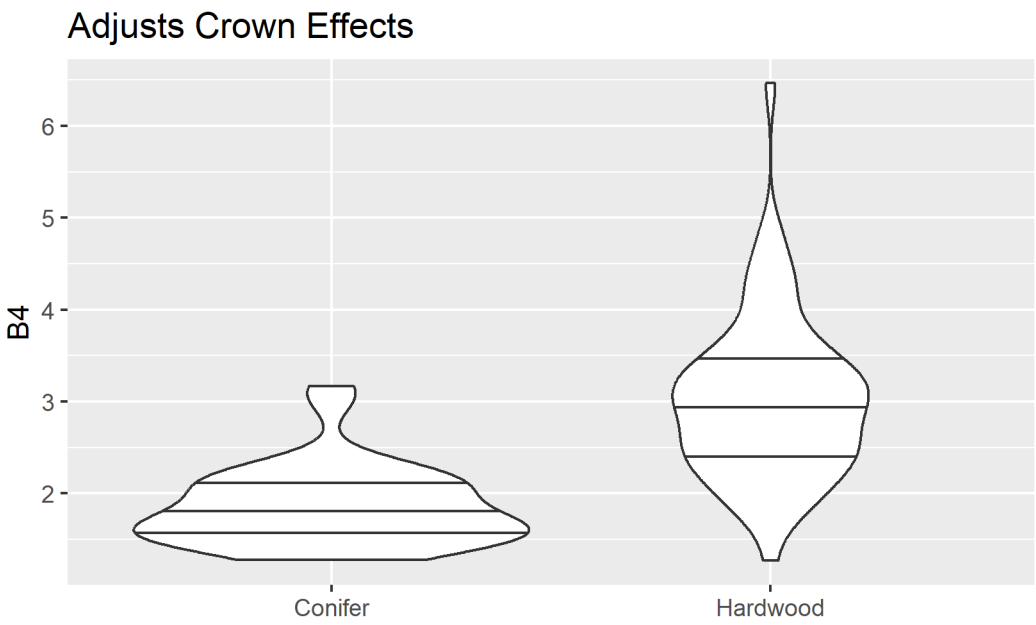
Equation 1 was fit to each qualifying species using an integrated fitting approach and errors minimized were ending `dbh`. 94 species equations were estimated. Sample sizes ranged from 5134 to 342704 observations.

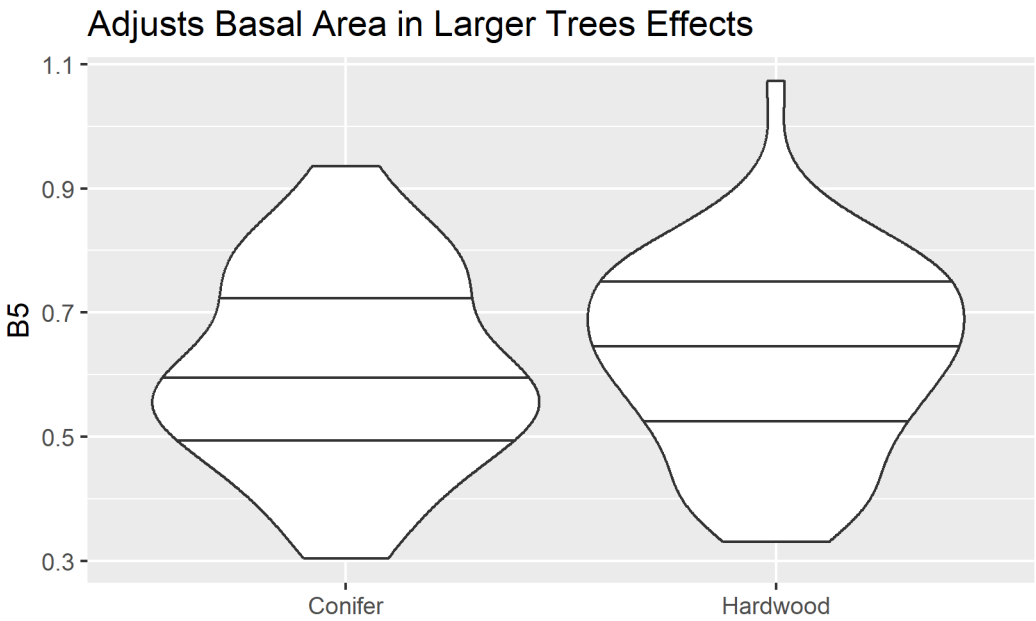
The graphs below show the range of parameter estimates divided into Conifer and Hardwood groups. The parameter estimates are remarkable stable across species.

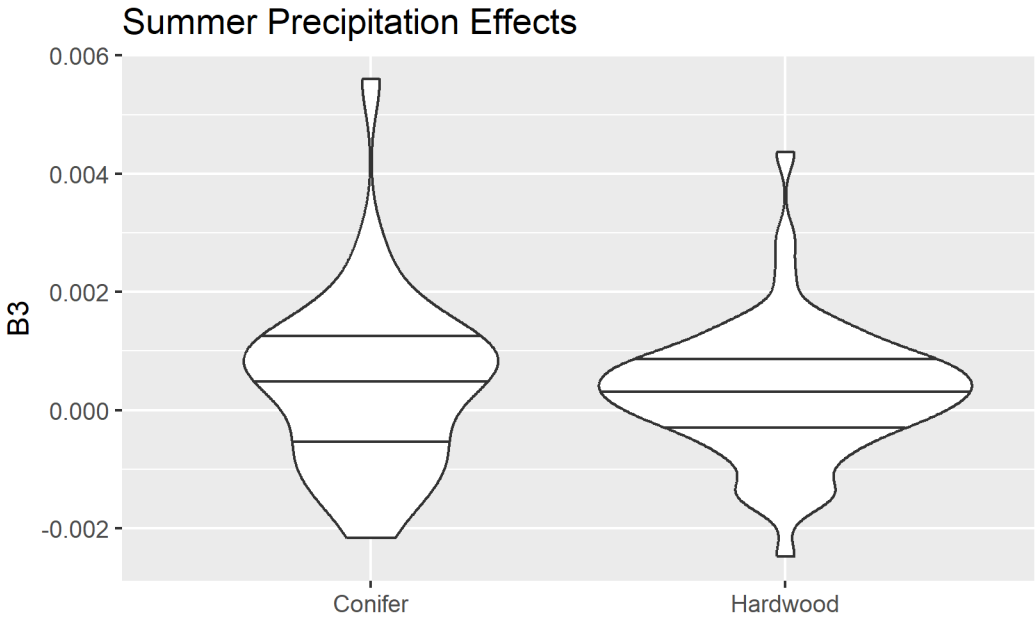

Sets Approximate Maximum DG
B0
0
-1
-2
-3
-4
Conifer
Hardwood

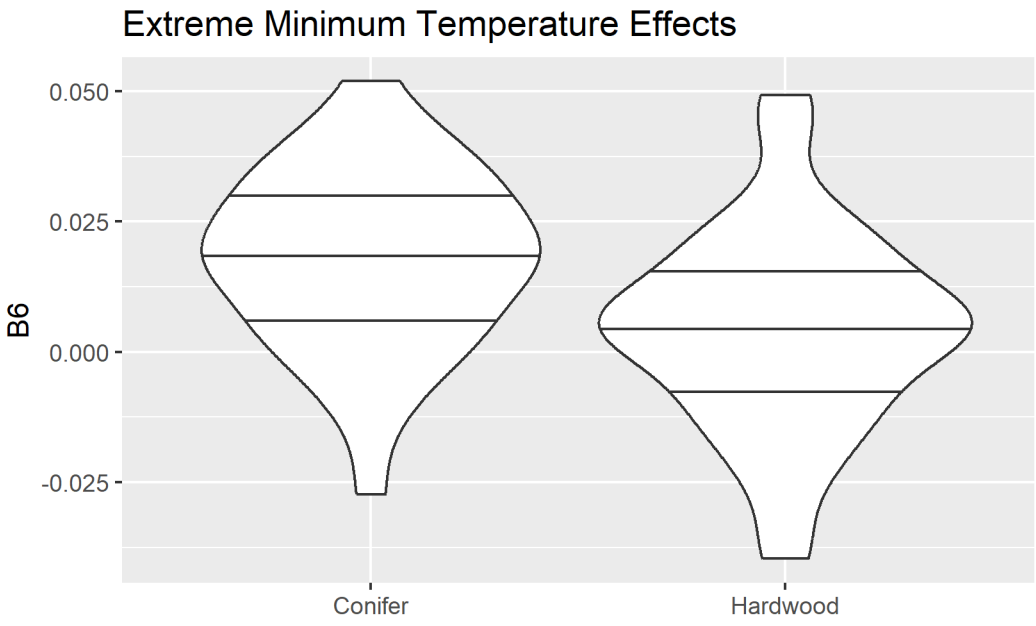

Sensitivity to Tree Size Relative to Crown
B1
-0.25
-0.50
-0.75
Conifer
Hardwood

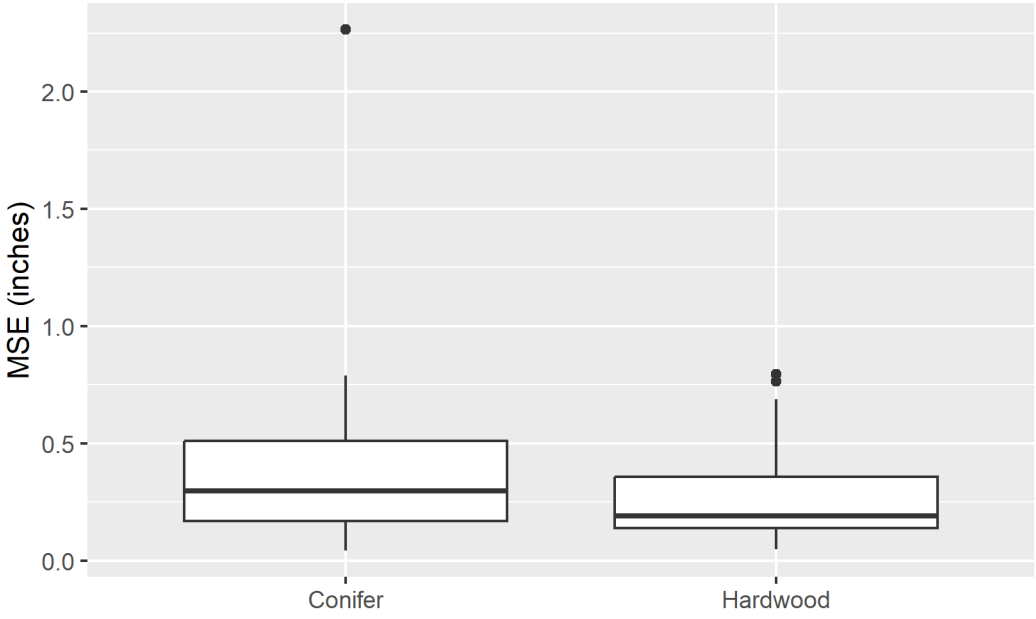

Sensitivity to Basal Area in Larger Trees Relative to DBH
B2
0.0
-0.1
-0.2
-0.3
-0.4
Conifer
Hardwood


Adjusts Crown Effects
B4
6
5
4
3
2
Conifer
Hardwood


Adjusts Basal Area in Larger Trees Effects
B5
1.1
0.9
0.7
0.5
0.3
Conifer
Hardwood


Summer Precipitation Effects
B3
0.006
0.004
0.002
0.000
-0.002
Conifer
Hardwood


Extreme Minimum Temperature Effects
B6
0.050
0.025
0.000
-0.025
Conifer
Hardwood


MSE (inches)
2.0
1.5
1.0
0.5
0.0
Conifer
Hardwood

FIA Species Clusters by Diameter Growth Parameter Estimates

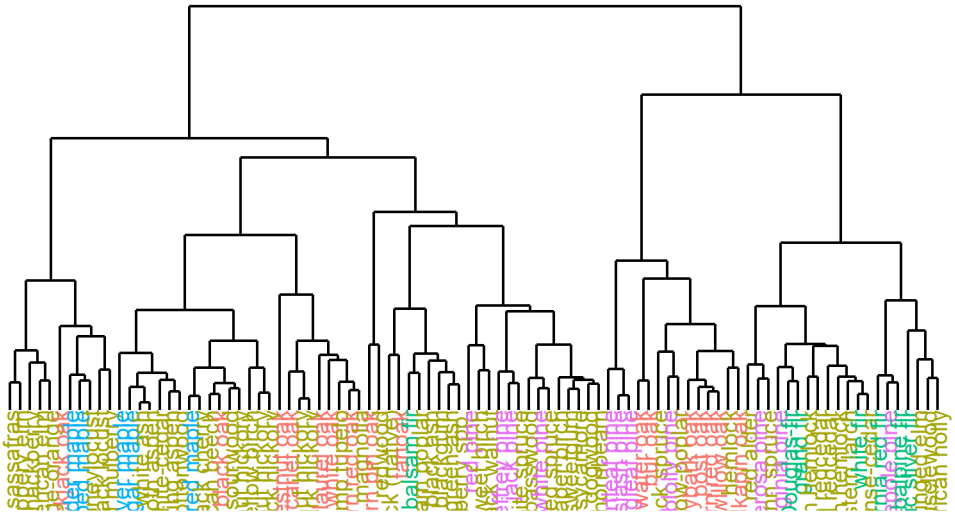


Table 1: Species with Negative Summer Precipitation Parameter

| FIA Species Code | Common Name |
|---|---|
| 131 | loblolly pine |
| 111 | slash pine |
| 621 | yellow-poplar |
| 827 | water oak |
| 110 | shortleaf pine |
| 108 | lodgepole pine |
| 835 | post oak |
| 832 | chestnut oak |
| 95 | black spruce |
| 68 | eastern redcedar |
| 261 | eastern hemlock |
| 19 | subalpine fir |
| 812 | southern red oak |
| 121 | longleaf pine |
| 820 | laurel oak |
| 711 | sourwood |
| 806 | scarlet oak |
| 71 | tamarack (native) |
| 403 | pignut hickory |
| 931 | sassafras |

| FIA Species Code | Common Name |
|---:|---|
| 823 | bur oak |
| 407 | shagbark hickory |
| 831 | willow oak |
| 602 | black walnut |
| 402 | bitternut hickory |
| 824 | blackjack oak |
| 631 | tanoak |
| 813 | cherrybark oak |
| 826 | chinkapin oak |
| 552 | honeylocust |
| 682 | red mulberry |
| 20 | California red fir |

Table 2: Species with Negative Extreme Minimum Temperature Parameter

| FIA Species Code | Common Name |
|---:|---|
| 131 | loblolly pine |
| 611 | sweetgum |
| 531 | American beech |
| 837 | black oak |
| 261 | eastern hemlock |
| 972 | American elm |
| 125 | red pine |
| 371 | yellow birch |
| 711 | sourwood |
| 409 | mockernut hickory |
| 105 | jack pine |
| 372 | sweet birch |
| 931 | sassafras |
| 313 | boxelder maple |
| 602 | black walnut |
| 462 | hackberry |
| 975 | slippery elm |
| 901 | black locust |
| 408 | black hickory |
| 402 | bitternut hickory |
| 824 | blackjack oak |
| 315 | striped maple |
| 826 | chinkapin oak |
| 641 | Osage-orange |

| FIA Species Code | Common Name |
|---:|---|
| 552 | honeylocust |
| 682 | red mulberry |
| 116 | Jeffrey pine |